

Lecture 1

Additive and Abelian Categories

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Conventions. Categories are locally small; $gf = g \circ f$. Rings are unital and modules are left modules. Biproduct inclusions and projections are denoted ι_i and ρ_i .

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1 Categories and Functors

Definition 1.1 (Category). A *category* $\mathcal{C}$ consists of the following data:

1. A class $\text{Ob}(\mathcal{C})$ of objects.
2. For every ordered pair of objects A, B , a set $\text{Mor}_{\mathcal{C}}(A, B)$ of morphisms with source A and target B . A member is written $f: A \rightarrow B$. Source and target are part of the data of a morphism.
3. For every three objects A, B, C , a composition function

$$\text{Mor}_{\mathcal{C}}(B, C) \times \text{Mor}_{\mathcal{C}}(A, B) \rightarrow \text{Mor}_{\mathcal{C}}(A, C), \quad (g, f) \mapsto g \circ f.$$

4. For every object A , a specified identity morphism $\text{id}_A \in \text{Mor}_{\mathcal{C}}(A, A)$.

The following diagrams are required to commute for every composable triple $A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$ (associativity) and every $f: A \rightarrow B$ (identity laws):

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & & A & \xrightarrow{\text{id}_A} & A & & A & \xrightarrow{f} & B \\ g \circ f \downarrow & & \downarrow h \circ g & & \searrow f & & \downarrow f & & \searrow f & & \downarrow \text{id}_B \\ C & \xrightarrow{h} & D & & & & B & & & & B. \end{array}$$

A diagram commutes when the composites along paths with the same endpoints agree. A category is *small* if its objects also form a set. Examples are sets with functions and $R\text{-Mod}$ with linear maps.

Definition 1.2 (Functor). A functor $F: \mathcal{C} \rightarrow \mathcal{B}$ assigns an object $F(A)$ to each object A and a function $\text{Mor}_{\mathcal{C}}(A, B) \rightarrow \text{Mor}_{\mathcal{B}}(F(A), F(B))$, written $f \mapsto F(f)$, to every pair. It preserves identities and composition: the following diagrams commute.

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ & \searrow F(g \circ f) & \downarrow F(g) \\ & & F(C) \end{array} \quad \begin{array}{ccc} F(A) & \xrightarrow{F(\text{id}_A)} & F(A) \\ & \searrow \text{id}_{F(A)} & \\ & & F(A) \end{array}$$

It is *faithful* if each of these functions is injective and *full* if each is surjective. A *full subcategory* is obtained by choosing some objects and retaining every morphism between them.

Definition 1.3 (Opposite Category). The opposite category $\mathcal{C}^{\text{op}}$ has the same objects and

$$\text{Mor}_{\mathcal{C}^{\text{op}}}(A, B) = \text{Mor}_{\mathcal{C}}(B, A).$$

Composition is reversed. A contravariant functor on $\mathcal{C}$ means a functor defined on $\mathcal{C}^{\text{op}}$. For example, $\text{Mor}_{\mathcal{C}}(A, -)$ is covariant, whereas $\text{Mor}_{\mathcal{C}}(-, B)$ is contravariant.

Definition 1.4 (Natural Transformation). For functors $F, G: \mathcal{C} \rightarrow \mathcal{B}$, a natural transformation $\eta: F \Rightarrow G$ consists of morphisms $\eta_A: F(A) \rightarrow G(A)$ such that the following square commutes for every $f: A \rightarrow B$:

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ \eta_A \downarrow & & \downarrow \eta_B \\ G(A) & \xrightarrow{G(f)} & G(B). \end{array}$$

A natural isomorphism is a natural transformation whose components are isomorphisms.

1.1 Isomorphisms, Monomorphisms, and Epimorphisms

Definition 1.5. A morphism $f: A \rightarrow B$ is:

1. an *isomorphism* if there is a morphism $g: B \rightarrow A$ with $gf = \text{id}_A$ and $fg = \text{id}_B$;
2. a *monomorphism* (or *mono*) if, for every object X and every $u, v: X \rightarrow A$, the equality $fu = fv$ implies $u = v$;
3. an *epimorphism* (or *epi*) if, for every object Y and every $u, v: B \rightarrow Y$, the equality $uf = vf$ implies $u = v$.

An inverse, when it exists, is unique: if g and h are inverses, then $g = g(fh) = (gf)h = h$.

Every isomorphism is both mono and epi, by cancellation with its inverse. The converse depends on the category.

Proposition 1.6 (Testing an Isomorphism by Maps into It). A morphism $f: A \rightarrow B$ is an isomorphism if and only if, for every object X , postcomposition induces a bijection

$$f_*: \text{Mor}_{\mathcal{C}}(X, A) \rightarrow \text{Mor}_{\mathcal{C}}(X, B).$$

Proof. An inverse to f induces an inverse to each f_* . Conversely, surjectivity for $X = B$ gives $g: B \rightarrow A$ with $fg = \text{id}_B$. Since $f(gf) = f = f \text{id}_A$, injectivity for $X = A$ gives $gf = \text{id}_A$. $\square$

2 Zero Objects, Products, and Coproducts

Definition 2.1. An object I is *initial* if $\text{Mor}_{\mathcal{C}}(I, X)$ is a singleton for every X . An object T is *terminal* if $\text{Mor}_{\mathcal{C}}(X, T)$ is a singleton for every X . A *zero object* is both initial and terminal and is denoted 0 .

Definition 2.2 (Zero Morphism). In a category with a zero object, the zero morphism $0_{A,B}: A \rightarrow B$ is the unique composite $A \rightarrow 0 \rightarrow B$. It does not depend on the choice of zero object. For $f: A \rightarrow B$, $g: X \rightarrow A$, and $h: B \rightarrow Y$,

$$f0_{X,A} = 0_{X,B}, \quad 0_{A,B}g = 0_{X,B}, \quad h0_{A,B} = 0_{A,Y}.$$

We usually write 0 when the source and target are clear.

Definition 2.3 (Product and Coproduct). A product of A_1, A_2 is an object P with projections $\rho_i: P \rightarrow A_i$ such that for every X and $f_i: X \rightarrow A_i$, there is a unique $u: X \rightarrow P$ making the left diagram commute. Dually, a coproduct is an object S with inclusions $\iota_i: A_i \rightarrow S$ such that for every Y and $g_i: A_i \rightarrow Y$, there is a unique $v: S \rightarrow Y$ making the right diagram commute:

$$\begin{array}{ccc} & X & \\ f_1 \swarrow & \downarrow u & \searrow f_2 \\ A_1 & \xleftarrow{\rho_1} P \xrightarrow{\rho_2} & A_2 \end{array} \qquad \begin{array}{ccc} A_1 & \xrightarrow{\iota_1} S \xleftarrow{\iota_2} & A_2 \\ g_1 \searrow & \downarrow v & \swarrow g_2 \\ & Y & \end{array}$$

Equivalently, the projections induce a bijection $\text{Mor}_{\mathcal{C}}(X, P) \rightarrow \text{Mor}_{\mathcal{C}}(X, A_1) \times \text{Mor}_{\mathcal{C}}(X, A_2)$ for every X ; the coproduct has the dual property.

Finite products include the empty product, a terminal object; finite coproducts include the empty coproduct, an initial object. Universal properties determine these objects up to unique isomorphism compatible with their structure maps.

3 Biproducts and the Addition of Morphisms

Definition 3.1 (Biproduct). In a category with zero morphisms, a biproduct $A_1 \oplus A_2$ is an object equipped with inclusions $\iota_i: A_i \rightarrow A_1 \oplus A_2$ and projections $\rho_i: A_1 \oplus A_2 \rightarrow A_i$ such that the projections exhibit a product, the inclusions exhibit a coproduct, and

$$\rho_i \iota_j = \begin{cases} \text{id}_{A_i} & i = j, \\ 0 & i \neq j. \end{cases}$$

A category is *semiadditive* if it has a zero object and all binary biproducts, hence all finite biproducts.

Theorem 3.2 (Canonical Addition). In a semiadditive category, each $\text{Mor}_{\mathcal{C}}(A, B)$ has a canonical commutative monoid structure. Its identity is the zero morphism, and composition preserves addition and zero in each variable.

Proof. The diagonal $\Delta_A: A \rightarrow A \oplus A$ and codiagonal $\nabla_B: B \oplus B \rightarrow B$ are the unique maps making these diagrams commute for $i = 1, 2$:

$$\begin{array}{ccc} A & \xrightarrow{\Delta_A} & A \oplus A \\ & \searrow \text{id}_A & \downarrow \rho_i \\ & & A \end{array} \qquad \begin{array}{ccc} B & \xrightarrow{\iota_i} & B \oplus B \\ & \searrow \text{id}_B & \downarrow \nabla_B \\ & & B. \end{array}$$

Given $f, g: A \rightarrow B$, let $s: A \oplus A \rightarrow B$ be the unique map whose restrictions to the two summands are f, g . Define

$$f + g = s\Delta_A = \nabla_B(f \oplus g)\Delta_A,$$

where $f \oplus g$ is uniquely specified by its diagonal components f, g and its two zero off-diagonal components.

If $g = 0$, then $s = f\rho_1$, so $f + 0 = f$; similarly $0 + f = f$. Interchanging the summands fixes Δ_A and exchanges the restrictions of s , proving commutativity.

Let $\Delta_A^{(3)}: A \rightarrow A \oplus A \oplus A$ have all three components id_A , and let $t: A \oplus A \oplus A \rightarrow B$ have restrictions f, g, h . Comparing components and restrictions shows that both $(f + g) + h$ and $f + (g + h)$ equal $t\Delta_A^{(3)}$.

For $k: B \rightarrow D$, the restrictions of ks are kf, kg , so $k(f + g) = kf + kg$. For $u: X \rightarrow A$, the restrictions of $s(u \oplus u)$ are fu, gu , and $\Delta_A u = (u \oplus u)\Delta_X$; hence $(f + g)u = fu + gu$. Composing a zero morphism with any morphism is zero. $\square$

Proposition 3.3 (Biproduct Identities and Matrices). In a semiadditive category,

$$\iota_1 \rho_1 + \iota_2 \rho_2 = \text{id}_{A_1 \oplus A_2}.$$

More generally a morphism $f: \bigoplus_j A_j \rightarrow \bigoplus_i B_i$ is uniquely determined by its entries $f_{ij} = \rho_i f \iota_j$, and

$$f = \sum_{i,j} \iota_i f_{ij} \rho_j, \quad (gf)_{kj} = \sum_i g_{ki} f_{ij}.$$

Proof. Composing the first asserted identity with either projection gives the corresponding projection, so it follows from uniqueness for products. Apply this identity to the source and target of f , then use bilinearity and $\rho_i \iota_j = \delta_{ij}$ to obtain both formulas. $\square$

3.1 Additive Categories and Functors

Definition 3.4. A category is *preadditive* if every morphism set is equipped with an abelian group structure and composition is bilinear. It is *additive* if it is preadditive and has finite biproducts. Equivalently, it is semiadditive and all its canonical morphism monoids are groups.

The structures agree: the map s used above is $f\rho_1 + g\rho_2$, so $s\Delta_A = f + g$ also for the given addition.

Definition 3.5. A functor $F: \mathcal{A} \rightarrow \mathcal{B}$ between preadditive categories is *additive* if every induced map on morphism groups is a group homomorphism. In particular, $F(0) = 0$ on morphisms and $F(f + g) = F(f) + F(g)$.

Between additive categories, an additive functor preserves finite biproducts, because it preserves the biproduct identities. Conversely, a functor preserving finite products or finite coproducts preserves zero objects and the biproduct structure, and hence the formula defining addition in Theorem 3.2. Thus it is additive. The functors $\text{Mor}_{\mathcal{A}}(M, -): \mathcal{A} \rightarrow \mathbf{Ab}$ and $\text{Mor}_{\mathcal{A}}(-, N): \mathcal{A}^{\text{op}} \rightarrow \mathbf{Ab}$ are additive.

4 Kernels, Cokernels, and Abelian Categories

Definition 4.1 (Kernel and Cokernel). In a category with zero morphisms, a kernel of $f: A \rightarrow B$ is a morphism $k: K \rightarrow A$ such that $fk = 0$ and, for every $u: X \rightarrow A$ with $fu = 0$, there is a unique $v: X \rightarrow K$ with $kv = u$.

A cokernel of f is a morphism $c: B \rightarrow Q$ such that $cf = 0$ and, for every $u: B \rightarrow Y$ with $uf = 0$, there is a unique $v: Q \rightarrow Y$ with $vc = u$.

We also write $\ker f = K$ and $\operatorname{coker} f = Q$ for the corresponding objects. For modules, $K = f^{-1}(0)$ and $Q = B/f(A)$.

Lemma 4.2. Kernels are monomorphisms and cokernels are epimorphisms. In an additive category, f is mono if and only if $fu = 0$ implies $u = 0$ for all u with target A . Dually, f is epi if and only if $uf = 0$ implies $u = 0$ for all u with source B . If a kernel of f exists, f is mono precisely when its kernel object is zero; dually for epis and cokernels.

Proof. If $kv = kw$ for a kernel k of f , both v, w are factorizations of the same morphism through k , so uniqueness gives $v = w$. The cokernel assertion is dual. In an additive category, $fu = fv$ is equivalent to $f(u - v) = 0$. This proves the cancellation criterion. If the kernel object is zero, every morphism killed by f is zero. Conversely, if f is mono, $fk = 0$ forces $k = 0$; since k is mono, $k \operatorname{id}_K = k0$ forces $\operatorname{id}_K = 0$, which makes K a zero object. $\square$

Definition 4.3 (Abelian Category). A category $\mathcal{A}$ is *abelian* if:

1. $\mathcal{A}$ is additive;
2. every morphism has a kernel and a cokernel;
3. every monomorphism is a kernel of some morphism, and every epimorphism is a cokernel of some morphism.

The definition is self-dual. Examples are **Ab**, **R-Mod**, and finite-dimensional vector spaces.

5 The Canonical Isomorphism from Coimage to Image

Lemma 5.1. In an abelian category, every mono is a kernel of its own cokernel, and every epi is a cokernel of its own kernel.

Proof. Let $m: M \rightarrow B$ be mono, so m is a kernel of some $g: B \rightarrow D$. Let $c: B \rightarrow Q$ be a cokernel of m . Since $gm = 0$, write $g = tc$. If $cu = 0$, then $gu = tcu = 0$, so u factors uniquely through m . This proves that m is a kernel of c . Reverse arrows for the epi statement. $\square$

Proposition 5.2. A morphism in an abelian category that is both mono and epi is an isomorphism.

Proof. If $f: A \rightarrow B$ is mono, it is a kernel of some $g: B \rightarrow D$. Since $gf = 0$ and f is epi, $g = 0$. The kernel property applied to id_B gives $s: B \rightarrow A$ with $fs = \operatorname{id}_B$. Monicity and $f(sf) = f$ then give $sf = \operatorname{id}_A$. $\square$

Lemma 5.3 (Pullbacks of Epimorphisms). If $e: E \rightarrow B$ is epi and $t: X \rightarrow B$ is any morphism in an abelian category, their pullback exists and its projection $p: P \rightarrow X$ is epi.

Proof. Put $r = e\rho_1 - t\rho_2: E \oplus X \rightarrow B$, and let $s: P \rightarrow E \oplus X$ be its kernel. Set $a = \rho_1 s$ and $p = \rho_2 s$. Then $ea = tp$. For maps $u: Y \rightarrow E$, $v: Y \rightarrow X$, let $b: Y \rightarrow E \oplus X$ be the unique map with components u, v . The equality $eu = tv$ is equivalent to $rb = 0$, so the kernel property gives a unique $Y \rightarrow P$ inducing u, v . This is the pullback universal property.

The map r is epi since $r1_1 = e$. By Lemma 5.1, it is a cokernel of s . If $w: X \rightarrow Y$ satisfies $wp = 0$, then $w\rho_2 s = 0$, so there is $z: B \rightarrow Y$ with $w\rho_2 = zr$. Restricting to E gives $ze = 0$, whence $z = 0$. Restricting to X now gives $w = 0$. The epi criterion in Lemma 4.2 proves that p is epi. $\square$

Definition 5.4 (Image, Coimage, and Comparison). For $f: A \rightarrow B$, choose its kernel $k: K \rightarrow A$ and cokernel $c: B \rightarrow Q$. Define the coimage and image together with their structure maps by

$$q: A \rightarrow \text{coim} f = \text{coker } k, \quad m: \text{im} f = \ker c \rightarrow B.$$

There is a unique morphism $\bar{f}: \text{coim} f \rightarrow \text{im} f$ such that $f = m\bar{f}q$:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ q \downarrow & & \uparrow m \\ \text{coim} f & \xrightarrow{\bar{f}} & \text{im} f. \end{array}$$

Indeed, $cf = 0$ first gives a unique $u: A \rightarrow \text{im} f$ with $mu = f$. Then $muk = fk = 0$ and monicity of m give $uk = 0$, so u factors uniquely through q .

Theorem 5.5 (First Isomorphism Theorem). In an abelian category the canonical morphism $\bar{f}: \text{coim} f \rightarrow \text{im} f$ is an isomorphism. In particular, every morphism factors as an epimorphism followed by a monomorphism.

Proof. We first prove that $\bar{f}$ is mono. Let $j: L \rightarrow \text{coim} f$ be its kernel, and form the pullback

$$\begin{array}{ccc} P & \xrightarrow{a} & A \\ p \downarrow & & \downarrow q \\ L & \xrightarrow{j} & \text{coim} f. \end{array}$$

The projection p is epi by Lemma 5.3. Since $qa = jp$ and $\bar{f}j = 0$,

$$fa = m\bar{f}qa = m\bar{f}jp = 0.$$

By the kernel property of k , write $a = kv$. Then $jp = qa = qkv = 0$. Since p is epi, $j = 0$. Because j is mono, its domain L is a zero object. Thus $\bar{f}$ has zero kernel and is mono.

Apply this same argument in $\mathcal{A}^{\text{op}}$. The coimage of the reversed morphism f^{op} is the image of f , its image is the coimage of f , and its comparison is the reversed arrow $\bar{f}$. The argument proves this reversed arrow mono, which says precisely that $\bar{f}$ is epi in $\mathcal{A}$. Proposition 5.2 now makes $\bar{f}$ an isomorphism. Hence $f = m(\bar{f}q)$ is an epi followed by a mono. $\square$

For modules, $\text{coim} f = A / \ker f$, $\text{im} f = f(A) \subseteq B$, and $\bar{f}(a + \ker f) = f(a)$.

6 Exact Sequences and Chain Complexes

Definition 6.1 (Exactness). For $A \xrightarrow{f} B \xrightarrow{g} C$ with $gf = 0$, there is a canonical morphism $\text{im} f \rightarrow \ker g$ compatible with their inclusions into B . The pair is *exact at B* if this map is an isomorphism. A sequence is exact if it is exact at each object where two adjacent maps are specified. A short exact sequence is an exact sequence

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0.$$

Equivalently, f is a kernel of g and g is a cokernel of f .

An additive functor is *left exact* if it preserves kernels, *right exact* if it preserves cokernels, and *exact* if it preserves short exact sequences.

Definition 6.2 (Chain Complex and Chain Map). A chain complex $C_\bullet$ in an abelian category consists of objects C_n , $n \in \mathbb{Z}$, and differentials $d_n^C: C_n \rightarrow C_{n-1}$ satisfying $d_{n-1}^C d_n^C = 0$. A chain map $f: C_\bullet \rightarrow D_\bullet$ consists of maps $f_n: C_n \rightarrow D_n$ making the following squares commute:

$$\begin{array}{ccc} C_n & \xrightarrow{d_n^C} & C_{n-1} \\ f_n \downarrow & & \downarrow f_{n-1} \\ D_n & \xrightarrow{d_n^D} & D_{n-1} \end{array} \quad (n \in \mathbb{Z}).$$

Composition and identities are defined degreewise; the resulting category is denoted $\mathbf{Ch}(\mathcal{A})$.

The cycle object is $Z_n(C) = \ker d_n^C$, and the boundary object is $B_n(C) = \operatorname{im} d_{n+1}^C$. The equation $d_n^C d_{n+1}^C = 0$ gives a canonical mono $B_n(C) \rightarrow Z_n(C)$. Define

$$H_n(C) = \operatorname{coker}(B_n(C) \rightarrow Z_n(C)).$$

A chain map induces maps on cycles and boundaries by their universal properties and therefore induces $H_n(f): H_n(C) \rightarrow H_n(D)$. The category $\mathbf{Ch}(\mathcal{A})$ is abelian, with kernels and cokernels computed degreewise; the induced differentials exist uniquely by the corresponding universal properties. A complex is *acyclic* if every $H_n(C)$ is zero, equivalently if its underlying sequence is exact in every degree.

Definition 6.3 (Chain Homotopy). For chain maps $f, g: C_\bullet \rightarrow D_\bullet$, a chain homotopy from g to f is a family $h_n: C_n \rightarrow D_{n+1}$ such that

$$f_n - g_n = d_{n+1}^D h_n + h_{n-1} d_n^C \quad (n \in \mathbb{Z}).$$

A complex is *contractible* if its identity chain map is chain homotopic to zero.

Proposition 6.4. Chain-homotopic maps induce the same maps on homology. In particular every contractible complex is acyclic.

Proof. Let $z: Z_n(C) \rightarrow C_n$ be the kernel inclusion. Since $d_n^C z = 0$, the homotopy identity restricts to $(f_n - g_n)z = d_{n+1}^D h_n z$. This factors through $B_n(D)$ and becomes zero in $H_n(D)$. Thus $H_n(f) = H_n(g)$. If $f = \operatorname{id}_C$ and $g = 0$, the identity of $H_n(C)$ is zero, so $H_n(C)$ is a zero object. $\square$

The converse fails for modules over a general ring. The exact complex of abelian groups

$$0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2 \rightarrow 0$$

(in degrees 2, 1, 0) is not contractible: a contracting homotopy in degree 0 would give a section $\mathbb{Z}/2 \rightarrow \mathbb{Z}$ of the quotient map, which is impossible. Over a field, every acyclic complex is contractible after choosing splittings of its short exact sequences of cycles.